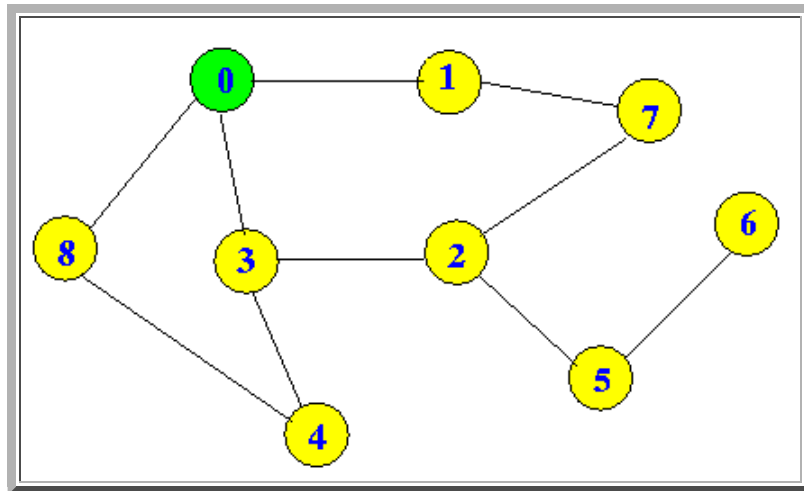
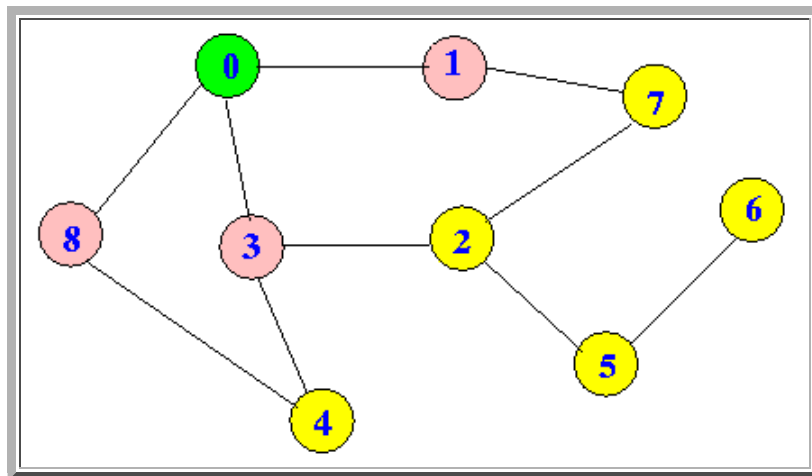

Graph traversal algorithms: *Breadth* First Search

- **Breadth First Search:** visit *all* your neighbors before your neighbor's neighbors
 - What is **breadth first**:

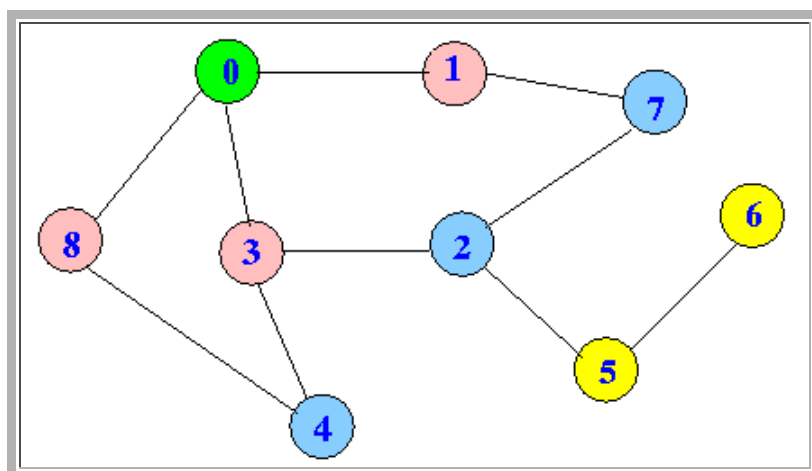
- Start at some node:



- Visit *all the neighbors* first:



- *Only* then visit the *neighbors' neighbors*:



- And so on

- **Data structure** that helps in the **implementation** of the **Breadth First Search** algorithm:

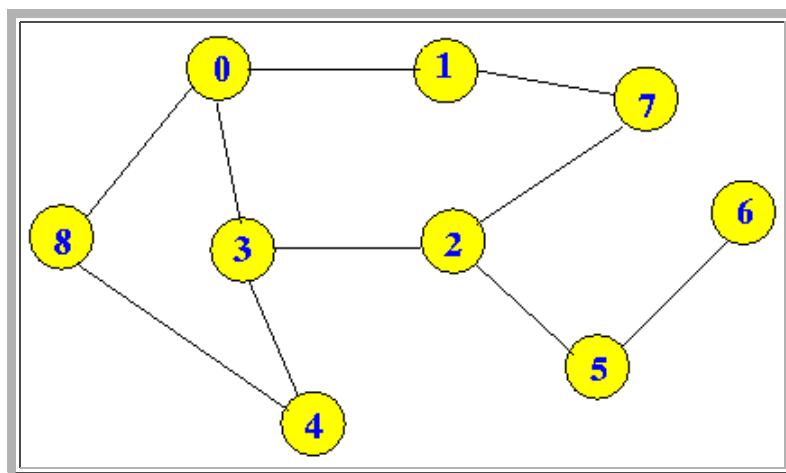
- A **queue**
- Nodes will be **visit** in the **order imposed** by the **FIFO queue**

- **Breadth first search:**

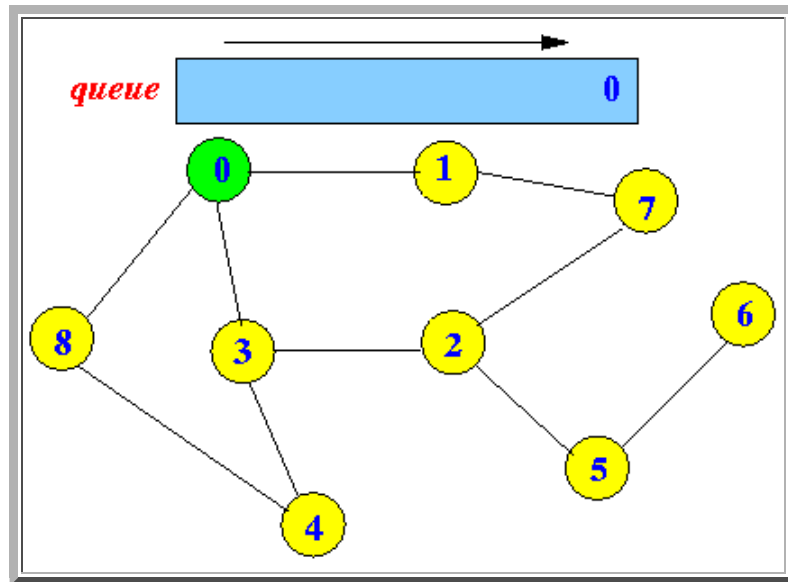
- **Put** an **arbitrary node x** in **queue Q**
Mark the node **x** as **visited**
- **Repeat:**
 - **h** = **remove** the **first node** from **queue Q**
 - Put **all of h 's unvisited neighbor nodes** in a **queue Q**
 - **Mark** all inserted nodes as **visited**

- **Example:**

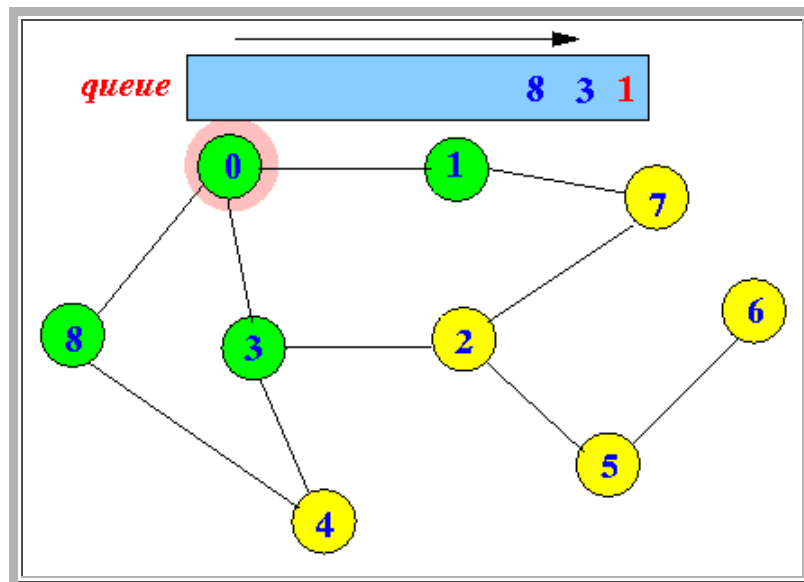
- **Graph:**



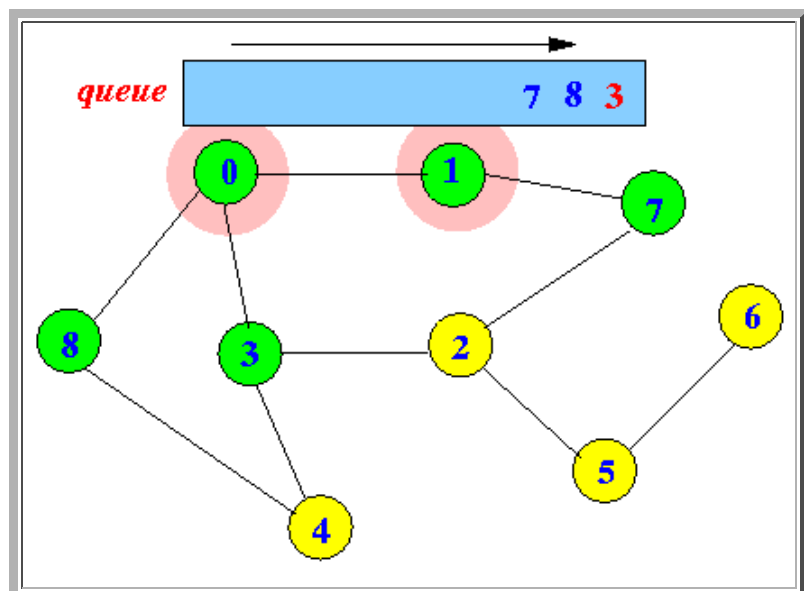
- **Initial state:**



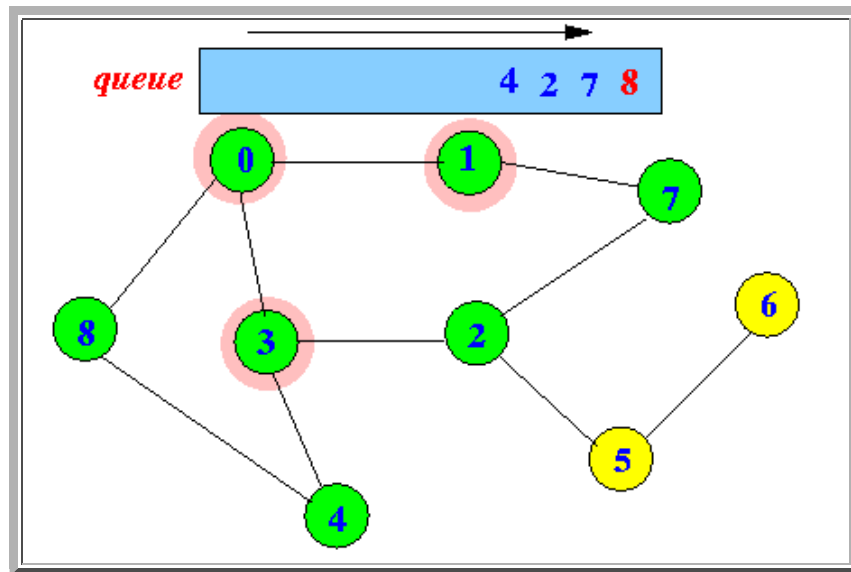
- State *after processing 0*



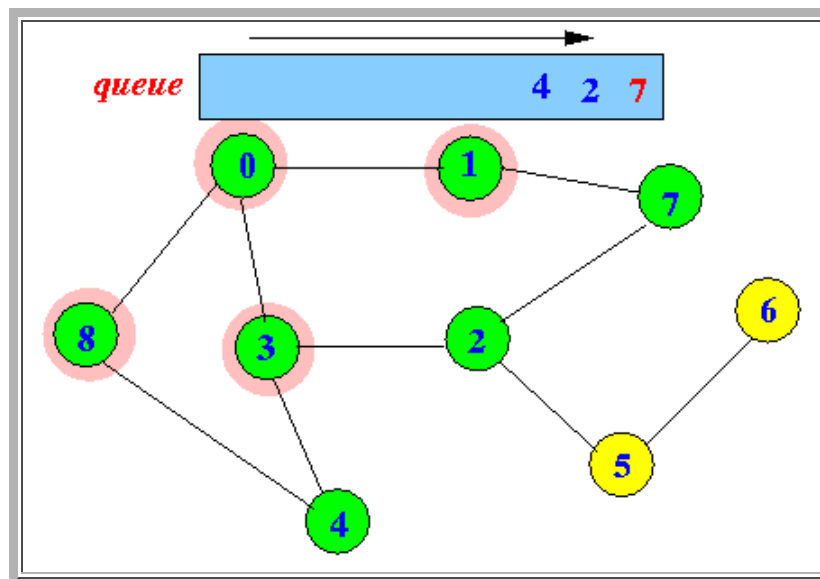
- State *after processing 1*



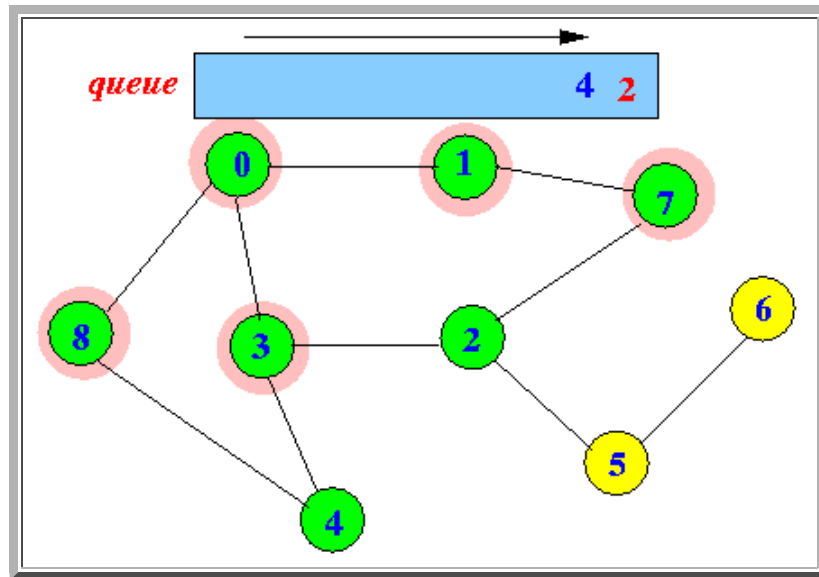
- State *after* processing **3**



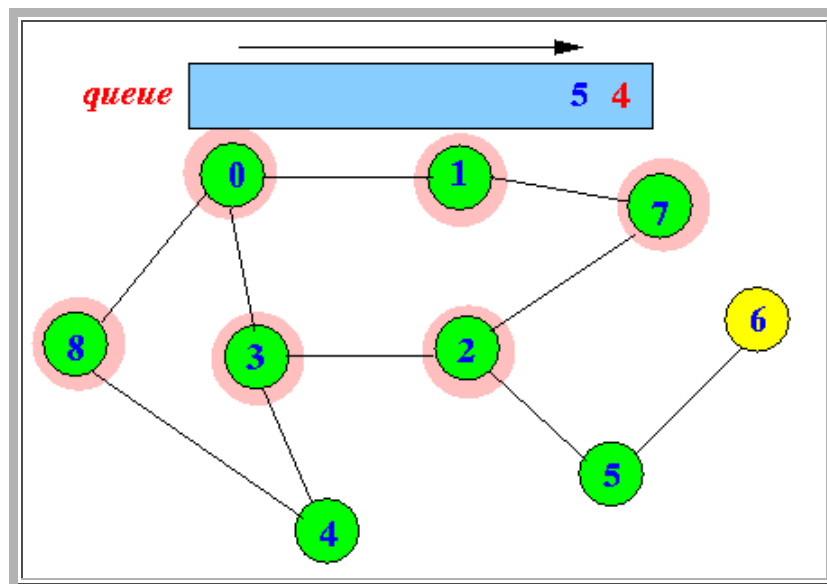
- State *after* processing **8**



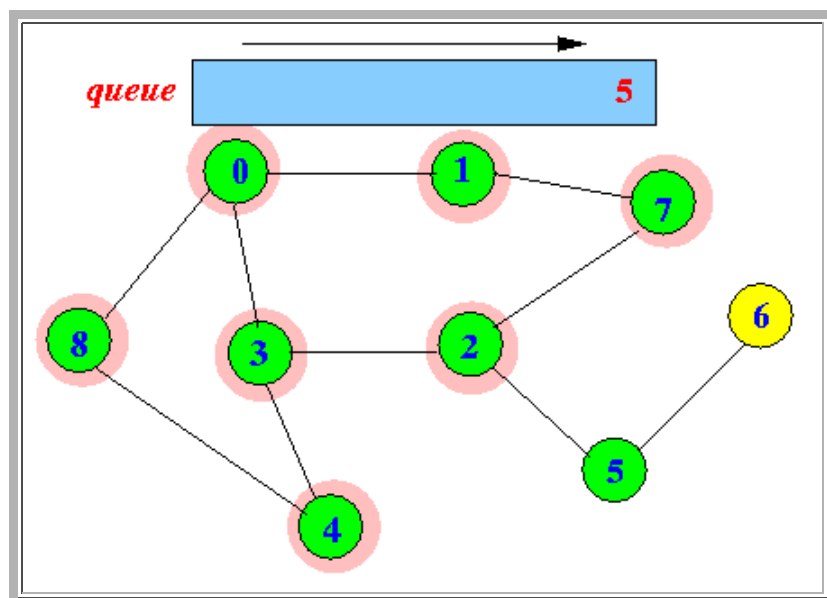
- State *after* processing **7**



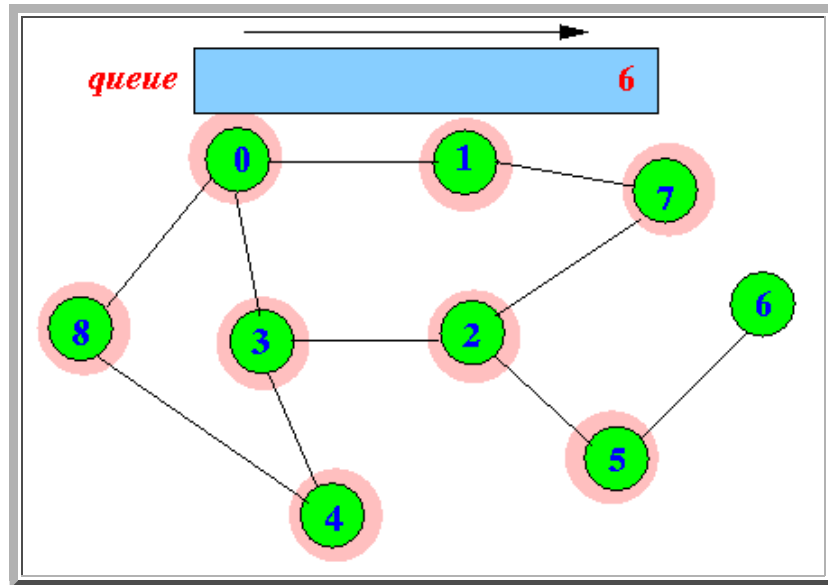
- State after processing 2



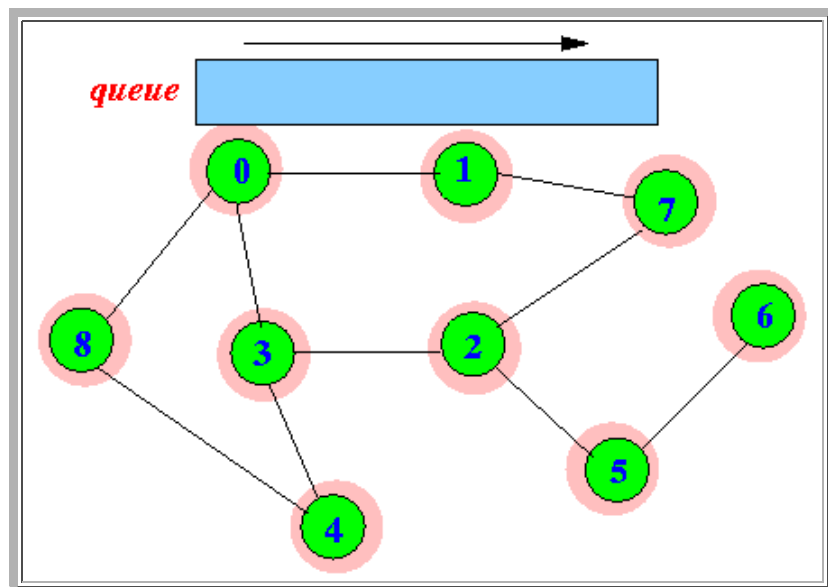
- State after processing 4



- State *after* processing **5**



- State *after* processing **6**



- **DONE**

(The **queue** has become **empty**)

- **Implementation in Java**

- **Java code:**

```
public void bfs()
{
    // BFS uses Queue data structure
```

```

Queue q = new LinkedList();

q.add(rootNode);
visited[rootNode] = true;

printNode(rootNode);

while( !q.isEmpty() )
{
    int n, child;

    n = (q.peek()).intValue();

    child = getUnvisitedChildNode(n);    // Returns -1 if no unvisited niode left

    if ( child != -1 )
    { // Found an unvisted node

        visited[child] = true;          // Mark as visited

        printNode(child);

        q.add(child);                  // Add to queue
    }
    else
    {
        q.remove();                    // Process next node
    }
}

```

- **Example Program:** (Demo above code)

Example

- The **BFS** Prog file: [click here](#)
- A **Test program**: [click here](#)